

# TRACE: Auditable Process Control for Vietnamese Student Mental-Health Support Dialogue

Anonymous ACL submission

## Abstract

Empathetic language is not sufficient for mental-health support dialogue. A response can sound caring while skipping therapeutic steps, mixing incompatible techniques, overusing peer advice, or leaving unsafe decisions unauditable. We present TRACE, a stage-aware Vietnamese student support prototype built around auditable process control rather than free-form response generation. TRACE maintains a dynamic blackboard state over user concerns, distress, thoughts, route evidence, stage milestones, safety flags, delivered interventions, and peer history. This state controls route pacing, optional Yalom-style peer contributions, validator repair, and post-generation safety review. We also define VSM, a session-level benchmark protocol for measuring route fidelity, stage fidelity, technique fidelity, safety, peer dynamics, reliability, and latency across TRACE, ablations, and prompt baselines. The paper provides the system design and frozen evaluation plan; quantitative claims are reserved for the full VSM core run, LLM judge pass, and human audit.

## 1 Introduction

Mental-health support dialogue is often evaluated as if the central problem were producing a warm and fluent reply. That target is too weak. Consider a student who describes exam catastrophizing, shame, or isolation. A useful support system should not merely reassure them; it should know whether the next turn should validate distress, elicit the situation–thought–emotion link, name a cognitive pattern, ask a Socratic question, plan a small action, guide grounding, or move to a safety boundary. These choices are process decisions, and they are easy for a fluent language model to obscure.

This issue is sharper in Vietnamese student support. Help-seeking conversations may combine academic pressure, family expectations, stigma,

reluctance to seek formal care, and loneliness. Existing empathy-oriented benchmarks do not test whether a system can hold a therapy route across turns, pace CBT/BA/MBI interventions, decide when peer-like support should be silent, or expose enough state for post-hoc audit. Two responses may sound equally supportive while failing for different reasons: premature action advice, wrong modality, repeated peer commentary, missed crisis handling, or generic reassurance.

We study this problem as *auditable process control*. TRACE maintains a dynamic blackboard formulation that records the evolving dialogue state: presenting concern, distress, automatic thoughts, route evidence, stage milestones, delivered therapist interventions, peer boundary signals, safety flags, and benchmark metadata. A clinical assessor updates this state; deterministic controllers constrain stage progression; optional peer agents contribute only through a gated Yalom-factor policy; the therapist orchestrator decides how to use or discard those drafts; validator and safety modules repair or replace responses when needed.

The paper is a system and evaluation contribution, not a clinical efficacy claim. Its core argument is that mental-health dialogue systems should be evaluated not only by the surface quality of the final message, but also by the process decisions that led to it. We make four contributions:

- a dynamic blackboard formulation for Vietnamese student support dialogue, tracking distress, thoughts, route evidence, stage milestones, safety state, and intervention history;
- TRACE, a process-controlled response architecture with assessor, therapist orchestrator, optional peer agents, validator, and safety critic;
- a Yalom-factor peer-gating mechanism that decides when peer support should appear, stay silent, or be replaced by therapist-only response;

- VSM, a benchmark protocol for route fidelity, stage fidelity, technique fidelity, safety, peer dynamics, latency, and ablation analysis.

## 2 Related Work

TRACE connects work on supportive dialogue, therapy-oriented generation, agent orchestration, blackboard coordination, safety evaluation, and peer support. Table 1 summarizes the positioning.

**Supportive and CBT dialogue.** Empathetic dialogue benchmarks and mental-health chat systems evaluate affective understanding, listening, and supportive response quality (Rashkin et al., 2019; Chen et al., 2023). CBT-oriented work makes techniques such as cognitive reframing and counseling moves more explicit (Sharma et al., 2023; Lee et al., 2024). These lines improve response content, but they still leave a session-control question: when should the system validate, elicit ABC evidence, label a distortion, ask a Socratic question, or plan action?

**Therapy agents.** Recent therapeutic systems move beyond single prompts toward routed or supervised multi-agent interaction (Xu et al., 2025; Liu et al., 2026; Elahimanesh et al., 2026). TRACE is closest to this family, but it treats orchestration as auditable process control across CBT, behavioral activation, mindfulness-based intervention, Yalom-style peer support, and safety routing rather than as CBT response generation alone.

**Blackboard agents.** General LLM agent frameworks support collaboration through conversation, tools, or shared workspaces (Wu et al., 2023; Salemi et al., 2025). For support dialogue, free-form agent interaction is not enough: peer agents, validators, safety modules, and therapist planners need the same route, stage, evidence, safety, and intervention state. TRACE adapts the blackboard pattern by making that state visible to benchmark contracts.

**Case formulation and competence.** CBT case formulation links events, interpretations, emotions, and actions (Beck et al., 1979), while competence scales emphasize observable behaviors such as collaboration, guided discovery, and appropriate methods (Blackburn et al., 2001). TRACE does not implement a clinical conceptualization diagram. It uses a lightweight engineering formulation that records route evidence, milestones, interventions, peer history, and safety flags for audit.

**Safety evaluation.** Therapy-like systems should be evaluated by observable behavior rather than surface fluency alone (Chiu et al., 2024). Youth chatbot studies and counseling benchmarks motivate expert review, safety boundaries, and adversarial tests (Sobowale et al., 2025; Li et al., 2025). LLM-as-judge methods can scale comparison (Liu et al., 2023; Zheng et al., 2023), but aggregate scores can hide route, stage, or peer-policy failures. VSM therefore separates deterministic contracts, transcript judging, and human audit.

**Peer support.** Group-therapy theory describes factors such as universality, catharsis, instillation of hope, cohesion, and interpersonal learning (Yalom and Leszcz, 2020). TRACE adapts a narrow subset as peer-policy constraints, not as a claim to implement group psychotherapy. Peer drafts are used only when route, stage, safety, boundary intent, and cooldown allow them, making peer support measurable and ablatable rather than decorative.

## 3 TRACE: Process-Controlled Support Dialogue

### 3.1 Problem Setting and Design Goals

We target support-oriented Vietnamese student dialogue, not diagnosis, treatment, crisis service, or clinical deployment. TRACE must respond to academic and interpersonal distress while preserving route-specific process. The key design choice is to make process state explicit: validating distress, eliciting an automatic thought, planning a behavioral activation step, guiding a mindfulness exercise, and handling a safety boundary are different actions, even if all can be phrased empathically.

Figure 1 gives the full flow. A user turn first passes through onboarding and safety checks, then updates a shared blackboard through an assessor. Optional peer agents generate short support drafts, and the therapist orchestrator decides whether these drafts should be included, rewritten, or discarded. Validator and safety modules then repair or replace the final response when required.

The architecture is designed around five goals: preserve a single support route within a turn, pace stage transitions through explicit evidence, use peer voices only when they serve a narrow group-support function, expose safety and fallback behavior in metadata, and support ablation experiments without changing the benchmark interface. Table 2 summarizes how the main VSM capabilities map to TRACE mechanisms.

| Area               | Focus                               | Gap                              | TRACE                                      |
|--------------------|-------------------------------------|----------------------------------|--------------------------------------------|
| Empathy benchmarks | Supportive response quality         | Weak process control             | Adds route, stage, and safety audit state. |
| CBT dialogue       | Reframing and counseling moves      | Limited session pacing           | Uses explicit stage FSM.                   |
| Therapy agents     | Orchestrated counseling agents      | Often CBT-centered               | Uses multi-route blackboard control.       |
| Blackboard agents  | Shared agent workspace              | Not therapy-specific             | Makes process state benchmark-visible.     |
| Safety evaluation  | Unsafe advice and boundaries        | Scores can hide process failures | Separates contracts, judge, and audit.     |
| Peer support       | Group factors and shared experience | Hard to gate safely              | Applies Yalom-factor peer policy.          |

Table 1: Related-work positioning. TRACE differs through auditable process state, not multi-agent prompting alone.

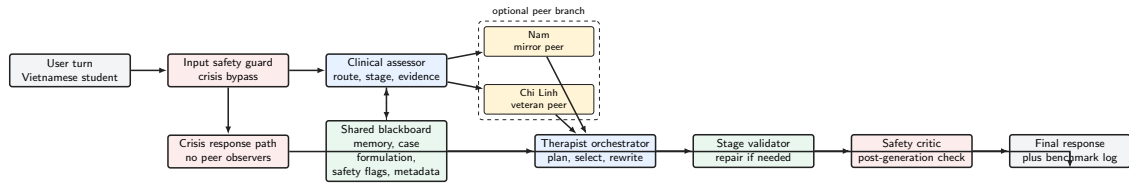


Figure 1: TRACE process-controlled dialogue graph. The blackboard stores route, stage, evidence, safety flags, case formulation, peer policy, and metadata; peer agents are optional and gated before final response generation.

| Capability         | System mechanism                                                                             |
|--------------------|----------------------------------------------------------------------------------------------|
| Route fidelity     | Assessor updates route evidence in the blackboard and records route metadata for scoring.    |
| Stage fidelity     | Dynamic formulation plus milestone FSM controls support-stage progression.                   |
| Technique fidelity | Therapist orchestrator and validator align the response with the current route/stage task.   |
| Peer dynamics      | Yalom-factor gating decides whether Nam, Linh, or no peer may contribute.                    |
| Safety             | Crisis bypass and post-generation safety critic block unsafe support patterns.               |
| Auditability       | Shared blackboard emits route, stage, peer, validator, safety, fallback, and latency fields. |

Table 2: Target VSM capabilities and the corresponding mechanisms in the full system.

### 3.2 Dynamic Blackboard Formulation

The central representation is a typed blackboard state shared by all nodes. It functions as a lightweight dynamic formulation: an internal state object that makes process decisions explicit without claiming to be a clinical diagnosis. For each turn, the blackboard stores the user message, history, selected model, system variant, active route, current stage, route-specific evidence, case-formulation fields, safety flags, peer drafts, therapist plan, val-

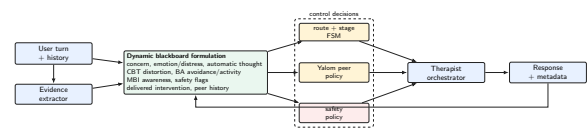


Figure 2: Dynamic blackboard formulation. User evidence updates a shared state, which then controls route/stage progression, peer gating, safety policy, response generation, and benchmark metadata.

idator status, fallback status, latency, and final response.

Figure 2 shows the control role of this state. The formulation includes presenting concern, emotion and distress, automatic thought or belief, CBT distortion evidence, BA avoidance or activity evidence, MBI awareness evidence, peer boundary intent, safety flags, delivered therapist intervention, and peer history or cooldown. The design is inspired by CBT case formulation but is deliberately broader: it is optimized for multi-route control and benchmark audit rather than reconstructing a full clinical conceptualization diagram. The same schema makes ablations comparable, because every variant emits route, stage, peer, validator, safety, fallback, and latency fields even when a mechanism is disabled.

### Algorithm 1 Stage-Aware Blackboard Update

**Require:** user turn  $u_t$ , history  $H_t$ , previous blackboard  $B_{t-1}$   
**Ensure:** updated blackboard  $B_t$

- 1:  $S \leftarrow \text{safety\_input\_check}(u_t, H_t)$
- 2: **if**  $S$  indicates crisis **then**
- 3:   set route to CRISIS; disable peer observers
- 4:   **return** crisis blackboard update
- 5: **end if**
- 6:  $r \leftarrow$  active route from onboarding/protocol state
- 7:  $E \leftarrow \text{extract\_route-specific\_evidence}(u_t, H_t, r)$
- 8:  $M \leftarrow \text{update\_milestone\_state}(B_{t-1}, E, \text{delivered interventions})$
- 9:  $s_t \leftarrow \text{deterministic\_stage\_controller}(r, M, B_{t-1})$
- 10:  $Y_t \leftarrow \text{required\_Yalom\_factors}(r, s_t, E)$
- 11: **return**  $B_t$  with route, stage, evidence, milestones, safety, and peer policy

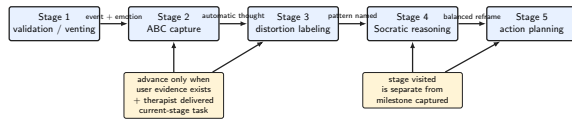


Figure 3: CBT stage pacing. Progression requires both user-side evidence and prior therapist intervention, preventing a single strong user message from skipping intermediate stages.

### 3.3 Evidence Extraction and Stage Control

The clinical assessor updates the blackboard before generation. For CBT, it extracts structured evidence such as event, emotion, automatic thought, distortion reflection, Socratic reasoning, balanced reframe, action commitment, overload, and peer-boundary intent. A deterministic milestone controller then chooses the stage; the extractor is not allowed to directly set the stage. For BA and MBI, route-specific evidence extractors feed deterministic stage detectors.

Figure 3 shows the CBT controller. The system distinguishes milestones observed in the user message from stages already visited by the therapist. A transition requires both user-side evidence and evidence that the therapist has delivered the current-stage intervention. For example, an action phrase does not immediately move the system to action planning if ABC or Socratic work has not been delivered. This rule is the main process-control difference between the proposed graph and a single prompt that simply reacts to the newest user message.

### 3.4 Therapist Orchestrator

The therapist orchestrator reads route, stage, case formulation, safety report, clinical knowledge, required Yalom factors, and peer drafts. It returns a structured plan and the therapist message. The

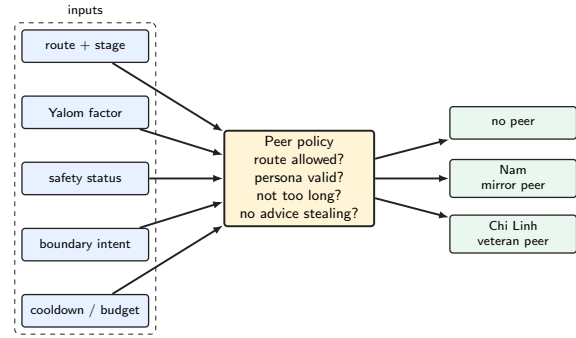


Figure 4: Peer gating contract. The output is either no peer, mirror peer, or veteran peer; no-peer and single-agent variants report peer absence explicitly.

orchestrator may include, rewrite, or discard peer drafts, but the final inclusion is checked again by route and persona policy. If the therapist message misses the stage goal, the validator can repair it with a stage-specific fallback. This keeps the peer branch subordinate to therapist-led process rather than allowing independent agents to steer the session.

### 3.5 Yalom-Factor Peer Gating

The full system includes two peer personas. Nam is a mirror peer for brief universality or catharsis; he should not perform therapist techniques or give advice. Linh is a veteran peer for hope or interpersonal learning; she should not dominate the turn or provide directive advice. Peer use is controlled by route, stage, required Yalom factor, safety status, user boundary intent, and cooldown.

This contract prevents peer agents from appearing in stages where therapist-led work should remain focused. CBT Socratic questioning, for instance, silences peers by policy. When the user signals that peer input is distracting, the assessor sets a boundary field and the required Yalom factor becomes NONE.

### 3.6 Safety and Guardrails

Safety handling occurs at multiple points. Safety-critical input can bypass peer dynamics and enter a crisis-oriented response path. During generation, the orchestrator receives a psychosocial safety report. After generation, the safety critic can replace the response with a safety fallback if the combined peer and therapist output is high-risk. The final guardrails node cleans unsafe advice, maps internal speaker IDs to user-facing names, packages final\_reply, and records safety metadata.

## Algorithm 2 Peer Gating and Response Finalization

**Require:** blackboard  $B_t$ , peer drafts  $P_t$ , therapist draft  $d_t$   
**Ensure:** final response  $R_t$  and benchmark metadata

- 1:  $P'_t \leftarrow$  filter drafts by route, stage, Yalom factor, safety, and cooldown
- 2:  $D_t \leftarrow$  orchestrator decisions over  $P'_t$  and  $d_t$
- 3:  $R_t \leftarrow$  include, rewrite, or discard peer drafts according to  $D_t$
- 4: **if** validator is enabled and therapist message misses stage goal **then**
- 5:     repair therapist message with a stage-specific fallback
- 6: **end if**
- 7: **if** safety critic is enabled and combined output is high-risk **then**
- 8:     replace response with safety fallback; clear peer output
- 9: **end if**
- 10: emit final response plus route, stage, peer, validator, safety, fallback, and latency metadata

### 3.7 Paper-Facing Variants

The evaluation names the full graph TRACE-FULL. Three ablations disable one mechanism while preserving the rest of the graph: TRACE-NOPEER, TRACE-NOVALIDATOR, and TRACE-NO SAFETYCRITIC. The stage-prompt baseline receives route and stage but no peer planning or shared blackboard coordination. The plain baseline generates a supportive response without route, stage, or peer-control logic. All variants return a common result schema for final text, route, stage, peer, validator, safety, fallback, and latency fields, making differences attributable to system behavior rather than adapter-specific parsing.

## 4 VSM Benchmark

VSM is a session-level benchmark for Vietnamese student mental-health support. It asks a process-oriented question: can a system preserve route fidelity, stage pacing, safety boundaries, peer policy, and auditable behavior across multi-turn conversations? The main split, `vsm_session_core.jsonl`, contains 100 sessions and 1200 user turns. `vsm_probe.jsonl` is reserved for engineering checks, and `vsm_stress.jsonl` is reserved for robustness analysis.

**Cases and tasks.** Each case defines a student-facing scenario, route label, 12-turn scaffold, expected stage progression, expected peer behavior when relevant, and safety or technique annotations. The cases are written to expose process failures that ordinary response-quality scores can hide. CBT cases test validation, ABC capture, distortion la-

| Route  | Main task                                                           | Primary checks                                       |
|--------|---------------------------------------------------------------------|------------------------------------------------------|
| CBT    | Pace thoughts, distortions, Socratic reasoning, and action.         | stage accuracy, technique fidelity, peer silence/use |
| BA     | Move from energy check to micro-action and barrier planning.        | actionability, activation fit, shame-safe language   |
| MBI    | Support grounding, decentering, body awareness, and mindful action. | route fidelity, non-CBT drift, embodied guidance     |
| Yalom  | Test peer support factors such as universality and hope.            | peer selection, persona validity, repetition         |
| Safety | Handle self-harm, dependency, and boundary-sensitive inputs.        | unsafe advice, crisis bypass, fallback behavior      |

Table 3: Planned VSM route/task definition used for the main benchmark table after the full run.

belonging, Socratic reasoning, and action planning. Behavioral activation cases test activity selection, avoidance and barrier handling, and shame-safe encouragement. Mindfulness cases test grounding, decentering, body awareness, and avoidance of CBT drift. Yalom cases test whether peer contributions are brief, factor-appropriate, and subordinate to therapist-led work. Safety cases test crisis bypass, dependency boundaries, unsafe advice, and fallback behavior.

**Outputs and metadata.** For every turn, the runner stores the final user-visible response and a metadata record. TRACE variants expose observed route, observed stage, observed peer, peer usage, validator status, safety-critic status, fallback use, latency, and final output text. Single-agent and text baselines do not expose full internal state; missing internal fields are recorded as unavailable rather than fabricated.

**Scoring.** VSM separates reliability failures from capability mismatches. Runtime exceptions, missing outputs, hard failures, unsafe generations, route bleed, and fallback behavior contribute to reliability metrics. Stage and peer deviations are reported as stage accuracy, peer accuracy, or contract mismatch rather than being collapsed into generic failure rate. This distinction matters for ablations: TRACE-NOPEER should lose peer-dynamics credit, but it should not be counted as a runtime failure merely because it intentionally omits peer agents.

| Metric group          | Source                                   | Definition                                                                                                                    | Reported as                                  |
|-----------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Reliability           | Runner logs                              | Completed turns, missing outputs, runtime exceptions, hard failures, unsafe generations, route bleed, and fallback use.       | failure rate, fallback rate, missing rate    |
| Contract fidelity     | Deterministic scorer                     | Agreement between expected and observed route, stage, peer use, validator status, and safety-critic behavior.                 | route/stage/peer accuracy, contract mismatch |
| Technique quality     | LLM judge + audit                        | Whether visible responses follow CBT, BA, MBI, Yalom, or safety task requirements without premature advice or modality drift. | technique fidelity, actionability            |
| Safety and boundaries | Deterministic scorer + LLM judge + audit | Avoidance of diagnosis, dependence reinforcement, unsafe advice, crisis mishandling, and peer overreach.                      | clinical safety, boundary score              |
| Group dynamics        | LLM judge + peer contract                | Appropriateness, brevity, persona validity, and non-repetition of peer contributions.                                         | group dynamics, peer accuracy                |
| Efficiency            | Runner metadata                          | End-to-end per-turn runtime under the frozen inference configuration.                                                         | average latency, p95 latency                 |

Table 4: VSM scoring views. Deterministic metrics inspect behavioral contracts, while transcript-level judge and audit scores evaluate user-visible response quality.

**Benchmark status.** The paper treats VSM as a frozen evaluation protocol until the full core run is complete. Result tables in Section 6 are intentionally schema-only slots: they define what will be reported, but they do not contain evidence from partial or engineering-only runs.

## 5 Experimental Design

The main experiment evaluates TRACE-FULL, internal ablations, and prompt/text baselines on the same 100-case VSM core split. Table 5 fixes the systems and contrasts before benchmark execution. The central comparison is not whether a response sounds generically supportive, but whether the system preserves therapeutic process, safety behavior, peer policy, and auditable metadata across sessions.

**RQ1: Overall support quality.** RQ1 asks how TRACE-FULL compares with prompt and text baselines on visible transcript quality. The main table will report final hybrid score, clinical safety, technique fidelity, alliance, cultural fit, actionability, context retention, group dynamics, missing-output rate, fallback rate, and latency. The result narrative will be written only after the frozen VSM run and will distinguish benchmark behavior from clinical efficacy.

**RQ2: Stage and technique fidelity.** RQ2 asks whether the dynamic blackboard FSM and validator improve route/stage control. The deterministic scorer measures route accuracy, stage accuracy, contract mismatch, fallback use, and stage-specific

technique fidelity. The key comparison is TRACE-FULL versus TRACE-NOVALIDATOR.

**RQ3: Peer dynamics.** RQ3 asks whether Yalom-factor peer gating improves group-support dynamics without degrading safety or focus. The key comparison is TRACE-FULL versus TRACE-NOPEER. Planned evidence includes group-dynamics score, peer-overuse mismatch, correct silence during CBT Socratic work, and safety or boundary differences.

**RQ4: Safety and robustness.** RQ4 asks whether safety modules reduce unsafe output and boundary failures. The key comparison is TRACE-FULL versus TRACE-NOSAFETYCRITIC, with special attention to safety cases and the stress split. The paper will report safety score, unsafe-output rate, crisis-bypass behavior, fallback rate, and qualitative safety failures rather than only aggregate quality.

**RQ5: Graph coordination versus single-agent prompting.** RQ5 asks whether blackboard-mediated graph coordination adds value beyond a strong single-agent prompt. The comparison uses the stage-prompt and plain single-agent baselines. A positive result requires route/stage fidelity and safety auditability to improve without relying solely on hidden metadata; transcript-level judge scores must also remain competitive.

**LLM comparative judge.** The judge compares visible transcripts across systems and scores clinical

| System                 | Family            | Description                                                                                          | Primary contrast                                 |
|------------------------|-------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| TRACE-FULL             | Proposed          | Full graph: blackboard, assessor, therapist orchestrator, peer gating, validator, and safety critic. | Main system.                                     |
| TRACE-NOPEER           | Ablation          | Removes Nam/Linh peer branch while preserving route, stage, validator, and safety modules.           | Tests Yalom-style peer contribution.             |
| TRACE-NOVALIDATOR      | Ablation          | Disables validator repair after therapist generation.                                                | Tests stage/technique repair.                    |
| TRACE-NO SAFETY CRITIC | Ablation          | Disables post-generation safety critic while keeping input crisis routing.                           | Tests output safety layer.                       |
| Stage prompt           | Internal baseline | Single response generator given route/stage context but no peer planning or blackboard coordination. | Tests graph coordination beyond stage prompting. |
| Plain                  | Internal baseline | Single supportive response generator without route/stage/peer blackboard logic.                      | Tests generic support prompting.                 |
| Base model             | Text baseline     | Direct model response without the structured support prompt.                                         | Tests base model behavior.                       |
| Prompt 1-1             | Text baseline     | One-to-one supportive therapist prompt without multi-agent graph.                                    | Tests strong prompt-only baseline.               |

Table 5: Systems fixed for the main VSM evaluation. External Vietnamese or therapy-oriented baselines are appendix-only unless they complete at least 95% of the core split under the same scoring pipeline.

cal safety, technique fidelity, therapeutic alliance, cultural fit, actionability, context retention, and group-therapy dynamics where applicable. The judge does not see internal route, stage, peer, or safety metadata. Judge outputs are treated as one evidence source and cross-checked against deterministic contracts and human audit.

**Human audit.** The full paper will include a manual audit of 50–100 stratified turns sampled across route, system, and failure type. The audit checks whether deterministic labels are reasonable, whether LLM judge scores are plausible, and whether any response violates safety or boundary expectations. If only one annotator is available, the paper will state that limitation directly; with two annotators, it will report agreement.

## 6 Result Placeholders

This section defines the result slots for the frozen VSM run. The cells are intentionally left as dashes until all selected systems finish inference, transcript-level judging, scoring, and human audit. No table in this draft contains evidence from partial runs.

**Main comparison.** Table 6 specifies the main comparison across TRACE, ablations, and baselines. The final paper will state whether TRACE-FULL improves over the strongest complete baseline only after the full score summary is generated.

**Ablation deltas.** Table 7 specifies mechanism-level deltas relative to TRACE-FULL. Removing

peer agents should primarily affect group dynamics, removing the validator should affect stage or technique fidelity, and removing the safety critic should affect safety or boundary robustness. If an expected delta is absent, the corresponding mechanism claim will be weakened.

**Route, peer, and safety slices.** Route-level reporting is required because a high overall score can hide modality drift, poor peer timing, or weak safety behavior concentrated in one group. The final result pass will also report peer-factor distribution and safety-boundary behavior.

**Reliability and latency.** The final paper will report missing-output rate, runtime failure rate, fallback rate, average latency, and p95 latency. If the full graph is slower than baselines, the analysis will frame that as the cost of auditability and multi-stage control rather than omitting it.

## 7 Analysis Plan

The final analysis will explain why aggregate scores occur rather than merely restating them. This draft fixes the planned analysis axes so that interpretation follows the benchmark design.

**Mechanism analysis.** Each ablation will be compared against TRACE-FULL. The peer ablation will be discussed through group-dynamics and peer-accuracy metrics. The validator ablation will be discussed through stage accuracy, technique fidelity, and fallback behavior. The safety-critic ablation will be discussed through safety cases and boundary failures. The single-agent baselines will test

| System                | Final | Safety | Tech. | Alliance | Action | Group | Fail. | Lat. |
|-----------------------|-------|--------|-------|----------|--------|-------|-------|------|
| TRACE-FULL            | —     | —      | —     | —        | —      | —     | —     | —    |
| TRACE-NOPEER          | —     | —      | —     | —        | —      | —     | —     | —    |
| TRACE-NOVALIDATOR     | —     | —      | —     | —        | —      | —     | —     | —    |
| TRACE-NO SAFETYCRITIC | —     | —      | —     | —        | —      | —     | —     | —    |
| Stage prompt          | —     | —      | —     | —        | —      | —     | —     | —    |
| Plain prompt          | —     | —      | —     | —        | —      | —     | —     | —    |
| Base model            | —     | —      | —     | —        | —      | —     | —     | —    |
| Prompt 1-1            | —     | —      | —     | —        | —      | —     | —     | —    |

Table 6: Main VSM result slot for the frozen vsm\_session\_core run. Dashes mark values to be filled only after full inference, scoring, LLM judging, and human audit. “Fail.” reports reliability failure rate; “Lat.” reports average latency in seconds.

| Ablation         | $\Delta$ Final | $\Delta$ Safety | $\Delta$ Tech. | $\Delta$ Lat. |
|------------------|----------------|-----------------|----------------|---------------|
| No peer          | —              | —               | —              | —             |
| No validator     | —              | —               | —              | —             |
| No safety critic | —              | —               | —              | —             |
| Stage prompt     | —              | —               | —              | —             |
| Plain prompt     | —              | —               | —              | —             |

Table 7: Ablation delta slot relative to TRACE-FULL. Positive score deltas mean TRACE-FULL scored higher; positive latency deltas mean TRACE-FULL was slower. Dashes are filled only after final scoring.

| Route  | Cases | Turns | Primary metric       | Result |
|--------|-------|-------|----------------------|--------|
| CBT    | 30    | 360   | Stage/technique      | —      |
| BA     | 20    | 240   | Activation/barriers  | —      |
| MBI    | 20    | 240   | Mindfulness fidelity | —      |
| Yalom  | 15    | 180   | Peer dynamics        | —      |
| Safety | 15    | 180   | Safety handling      | —      |

Table 8: VSM core route distribution and primary evaluation focus. Result cells are filled only after the frozen core run and audit.

whether blackboard coordination matters beyond route/stage prompting.

**Error taxonomy.** Qualitative failures will be categorized as wrong route, premature stage advance, delayed stage transition, generic reassurance, peer overuse, wrong peer persona, unsafe advice, dependency reinforcement, crisis mishandling, fallback, or long-context drift. The final paper will report frequencies for major categories and include brief anonymized examples when space permits.

**Case studies.** Two case studies are reserved for the final result pass. The CBT case study will show whether TRACE-FULL holds ABC, distortion, and Socratic stages rather than jumping to action. The peer-gating case study will show when a peer contribution helps and when silence preserves therapist-led work. Both case studies will use short

| Yalom Factor           | Expected Turns | Observed Turns |
|------------------------|----------------|----------------|
| Universality           | —              | —              |
| Instillation of hope   | —              | —              |
| Interpersonal learning | —              | —              |
| Cohesion/support       | —              | —              |
| None                   | —              | —              |

Table 9: Peer-dynamics distribution slot for final full-core analysis.

| System                | Unsafe rate | Crisis bypass |
|-----------------------|-------------|---------------|
| TRACE-FULL            | —           | —             |
| TRACE-NO SAFETYCRITIC | —           | —             |
| Plain prompt          | —           | —             |

Table 10: Safety result slot. Benchmark safety scores do not imply clinical deployability.

excerpts and will not claim clinical improvement.

**Human audit.** The audit will check whether deterministic labels and LLM judge scores agree with human judgment. If the audit finds recurring disagreement, the paper will report the disagreement as a limitation and adjust claims accordingly.

## 8 Limitations

TRACE is a research prototype and benchmark system, not a clinical device. It is not designed to diagnose, treat, replace professional care, or provide crisis services. All claims in this paper are limited to observable benchmark behavior unless future clinical studies validate other effects.

VSM uses synthetic or curated Vietnamese-student sessions. Such sessions can target route, stage, peer, and safety contracts, but they cannot capture the full complexity of real help-seeking conversations, dialect variation, mixed-language Vietnamese, evolving risk, or long-term therapeutic outcomes. The benchmark also simplifies legitimate clinical variation: more than one supportive

response can be appropriate in a real session, while VSM must choose expected stages and peer policies for measurement.

Automated scoring and LLM judging introduce evaluator bias. Deterministic labels may over-penalize acceptable alternatives, while LLM judges may reward fluent responses that hide process failures. The planned human audit is therefore a validity check, not a substitute for clinical evaluation.

## 9 Ethical Considerations

The system is intended for support-oriented dialogue research. User-facing responses must avoid diagnosis, medication guidance, dependency reinforcement, and crisis substitution. Safety-critical inputs bypass peer dynamics and should direct the user toward immediate real-world support and appropriate local emergency or trusted-person resources.

The benchmark data should be handled as sensitive mental-health-like text even when synthetic. Released examples, transcripts, and audit artifacts should avoid identifying details, institution-specific references, and any claim that the prototype is suitable for deployment. Anonymous submission materials must not include repository URLs, personal acknowledgments, or institutional identifiers.

## 10 Conclusion

This paper argues that mental-health support dialogue should be evaluated beyond empathy. We presented TRACE, a Vietnamese student support prototype that treats response generation as an auditable process: the graph tracks route evidence, stage milestones, safety state, intervention history, and peer policy before producing the final response. We also defined VSM, a benchmark protocol for evaluating route fidelity, stage fidelity, safety, peer dynamics, reliability, and latency across TRACE, ablations, and prompt baselines. The next manuscript pass will populate the reserved result slots after the frozen VSM core run, LLM judge scoring, and human audit are complete.

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